

A Packing of 68 Lines in $\mathbb{R}^{15}$

Limin Ge*

October 8, 2026

Abstract

We give 68 1-dimensional subspaces of $\mathbb{R}^{15}$ whose smallest squared chordal distance is at least 0.923303. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.922621, obtained by deleting 1 of the 69 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 68 subspaces of dimension 1 in $\mathbb{R}^{15}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.923303\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 15$, $n = 1$. The cell $N = 68$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.922621, obtained by deleting 1 of the 69 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(15 - 1)/15 \cdot 68/67 = 0.947264$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m15n1N68.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 74 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
230825795089814512943971297733483032045538384066
denominator:
2499999999999938664678919705236634137734801134107

B Coordinates

68 blocks of 15 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m15n1N68.txt`.

248771041506	206665422843	-204947142602	238891742303
272239946666	-296654722752	-8457398775	340419147387
-329197077805	-78629548278	-345060364902	147454189794
78594513259	207664620852	-99961467945	-259624080957
-2692079075	-32073945783	121608651029	189161245831
-118025496012	2637344146	424108822729	-98295648329
-350150718318	183813938158	-113415132632	167804928407
-376940993267	331909124162	558677533837	366792291880
220080815733	-20991147362	84469873458	211789419106
279142999895	-277815606502	-260691184778	164651092187
-165568933435	77053009571	-41096589560	-272724283992
316670241882	472266175329	98172716212	-17043520175
-187591718712	-134897316926	119158944804	-241309832227
415571353427	-321943755244	167177453610	390813971964
94252678903	261382154775	87253414127	414920829306
-424798809148	20023759118	586466126561	-30110900619
-332711325076	-168734603301	-102120971848	403664770249
437029851847	518356552304	-24045175986	214025263590
-368417288005	153124120670	250849198451	244008999812
-147947396909	195623721231	-311842251263	-48969199611
351173056110	198292056409	483492059293	318656968945
-192592741963	-304483970648	-225596740020	109963373735
-170141779106	65949035670	371927217254	-20277986632
-86275206564	-487796913350	24303957508	-118034289184
-122912067452	-309891069620	14614578606	-161579701622
-160137684035	1783229849	43436792105	-221779805242
73509695068	-12577766563	-298754161848	2830050793
233738725514	-48687474288	390595142577	172736904649
64046721014	-41841709865	-28493457081	40620126248
241944837877	317047400466	300472031025	-635511694153
323316895986	152748963158	-298893236342	398944035633
206530504375	37451978586	23969138625	-102516069929
103109121079	83183939380	-387102285179	166376140384
-100555151958	616465816575	218491872175	213532695062
144472924000	-114699575641	-201568848908	151796654366
77842101826	464401260739	209997915960	194270612074
-241056324404	-298251938989	3501297556	209437514636
156025364024	220688157438	329231027217	-160924925709
-355995789088	-102526119685	-387464888170	636350187249
-246302818460	55255270564	185572774719	-288563449730
392103702994	463444224587	-406672438514	-333305106941
262330032292	-29347452467	83997040228	-48878667183
237698605401	313495454104	-382868875249	92730527951
-505910799193	-291404444880	-51861112286	127339426548
-16667874304	1899916575	-155281574410	2560571872
319117789118	-55162107812	269211569064	-122410560385
-326653733262	-278240483689	-84522597650	74217419308
-24384067498	115584614155	212217098054	-81799223307
340818822788	154854394514	-101623726028	245658327438
-151009224565	341273414310	-339207186399	63171181516
22280055820	107167344770	-15475994022	-110734627137
-175085969542	231359718338	609858855287	221347958025
-112739732466	183312153488	54005264739	461594852512
-7595645701	-11346431510	-169577801702	-221474456698
140772773661	-240547649969	-16942422675	494126099607
-269927851636	398034124626	156665782386	-210490156704
8105539510	-281443661149	134140095693	-452353040295
-199703170604	-289346029133	191458973694	-38514892092
211779399728	-222802254626	100266881238	-197044121937
-656132843795	240018587290	-292210334722	227376606267
71991239210	380680500420	396770642339	-5044511225
123293478371	145285821050	260024843033	-98502128209
-20418412123	-104690528855	-191473608842	-35788483546
-343454934929	46882556501	-265224495101	38720201703
-78496583770	485382630022	-629502127671	28335329660
-489005061689	64200415400	-148857727722	-672777125738
-118064003903	313291130925	81989100609	-25134019499
28136197964	-230586560899	16747932518	233381829925
574378403458	-63606631789	-155888527661	-345380278349
-300430102701	273206165610	208767510625	-282692331117

112504070568	-204241416100	-313966354063	-42464106918
-240487574303	180129541567	130310086747	340475269651
384379841028	184011190035	417804599493	454180039029
87577203601	-431161408147	24719427744	409055995133
232054963230	65885286466	324320848370	-360424908172
-70717216158	407598181989	88167991766	137898245607
592008771050	14914747982	-405289702959	-11417266662
-245190973415	309574332255	57353179693	-329594816002
-306659591326	-127465006560	219237359393	192991550558
392831035571	401988743532	-394436520681	213022497630
42191830923	126757550293	-24299349632	5692472001
254308400785	164955531631	-142990429079	19715563553
8937875984	-201351270025	-176654170027	-80527500871
-254482225597	-305150355885	-29031030680	-291249506865
168329140153	-334624766971	-319937379587	-167652169244
-179039391744	60740373387	584877672368	-63665845095
239178748516	-315710756107	-58845312040	-350048735917
-66866801953	-436614242533	187569229224	-474530575272
-240689331802	-130518217296	-136564529193	3818973458
-157343403889	147621145573	-426573475001	-271635124116
145902552227	-435568709136	263613379367	-119267272895
336467529484	101222250814	24421185948	-334008072250
610544433470	-21433304160	-150255918874	-135887388565
305913633336	-68707973425	39145346469	-488904556002
261005093245	-146147516086	324034057866	-159842380433
-20932390473	-173397443971	254178986922	-85023404992
-295963332501	-294666941132	129980815659	-16289031660
29151731186	-199641141702	-202051987547	-354034227505
-110273158270	-144609730349	-385859049993	68282838115
-163680440104	-9019585397	21525475292	-65704612859
-178996893724	-438786239876	234841979762	201205624784
-70433249345	-85892750507	-22475787478	265441164089
342404411967	331137816254	211976202095	-127631814666
113798451517	38985676379	352860751092	90861214129
189750873203	550293603193	-527263027797	69959531588
-327561751823	347981923645	-298064845449	225402847709
41029277544	-188133034676	1108353288	435915795679
-119578847062	160571955275	37202978549	-411190646829
107484895088	-140067303017	-365974158628	-262179420766
82624488207	171937466700	-180880349605	341613822684
-167646355527	1624135070	181493428677	228911614232
-731581936131	372957185509	239276526941	36546723683
-125154904368	-23630192973	366076131260	454091853663
120437927150	-152214886818	341894051911	4448084925
-115834053605	-53052068954	405585686769	227138151615
245566119341	-612309453852	-177342779560	100544186220
146115702910	69483202091	-129008118528	-352620502078
300395895926	-417947002119	167421960708	249842160127
15494263015	-9938865468	-468740536626	-63508726170
281614455427	-449044679742	-80575802980	57451566925
-153664514882	36706237700	220068618191	-228064692458
-74897153933	165398682070	-220893195665	182837463848
-204898542806	17384525478	156311927228	266462096412
-33268415550	-32477373421	-205807580265	165618933540
379457708529	-450294960759	-239603155971	-136460803050
-120617188381	-320092682069	-59883606710	-458757964230
26604885710	-22001779304	-555578694219	448361892644
-120502139439	-482103803006	295790356975	357308409163
16753691815	22314988188	300807600844	50496868642
-45036563167	79999130799	84300457980	-215088769326
541185184530	8457730677	126650138587	-20506165687
-538218369378	-331059714011	15339140064	-98459515647
-293863467206	24621498397	-281306695856	165775281779
-287158700103	-455220552168	78543330612	-396817456508
15392755733	-242389534354	11050825353	-239894490362
-305322167278	273621859435	429742016434	185704651668
35342808933	-22350333634	-321655942118	174335434323
-152033045376	47845440084	245559175600	-46235465147
12064917981	75331778069	-186507918089	77628088990
-577595067785	-51373309095	-18397720823	364020992268
-74003834666	24033374235	144860918391	-434000043968
-42336922321	-358930537966	-198251580810	70218320434
309223859965	-406240979538	70544966721	261428706525
299113549561	-45252953730	387790442693	-483674824719
364273712726	-267518476133	-12561080124	42883186928
217412794840	-71919686090	-83028856619	-74514099527
146659480173	7459320492	-535455173735	-26300767387
92032455298	116963254703	94073751790	355983268061
139551625519	323905930948	-329552195136	327112796447
356200788382	-474259584689	-725467490	72173304522
-291683615255	366667256069	102569847703	-207490986283
-36212048730	-350435305002	197104944150	21639948023
445965061878	132800612636	185312301045	330205459714
266552056646	32346022288	-130617495692	202111497221
234965925449	-81494065108	403507090209	-80051305293
342066666713	231567848703	361287899154	523656687106
383823233742	5920084216	-168823977412	350134368449
-231143850876	207556523308	314417165704	-249943693312
49499635500	80782618732	-41890804596	-306406810243
50699897193	-334418150603	224377642276	-57196719150
58946273288	454832102278	-189272600063	133378768075
-227093081620	553917814888	221684725595	-509451828147
-267755169466	104569281719	449941253106	1851661577704
-20257620383	-36487635889	24137339124	416154510079
372207400976	-219806848612	-19094114047	388976412590
-52218657652	91024847431	216472854392	-94441570052
214483385615	-335825884623	549540661838	221667990043
-124526481707	286129039842	141303316778	114569559583
206039327723	-360408357994	-30252773448	301289570265
298058779812	134803711375	-392861905563	233280008581
-415870573939	18018893300	-89989856914	119429254446
266582666019	-275933661804	-276813210803	-110962718438

-23014266438	506756526518	-319810062527	-269900397605
342258608237	-22496754702	21214468785	-429116303090
266459553247	233688381044	-209107674153	430557188929
-20126956438	-127391136769	-304704777091	23330583551
121195097023	43619353024	551510225322	-136956793656
-312153672419	-206539796689	-119602276825	-116301170034
42567347840	-243548296732	-120106205831	-165718922102
-98633531242	-312189659117	139000916786	-99068101492
56287833270	205707863044	-364775775354	3232654288
-92971436380	-612433409640	-126587002188	-521907592774
284507043933	-156787059184	450646668808	-333482950137
123573867648	-33003822265	268130552398	49208446387
243151773515	86557828175	-100211650224	173644062858
-31833957942	77356020719	82009557818	465686159814
-35363449672	256661622933	128235011444	-110861817536
-436309306461	-20319286324	231278463589	-447894013123
-667192742381	-162316469497	-253060626818	-52126197365
471627664093	25639130620	24069696740	-188897507443
-385127830588	180636806208	-559268170625	-39111923874
105807301642	434920530788	295682771865	-105365660860
421594952002	247800120683	334156994926	-210374965603
113945805006	-56196983582	-131294181564	151367371313
218216791356	425058016790	-299698173458	189350560174
49838663173	-159570571672	-132847045871	43865526264
2582002258	210074298332	-365069160670	269561098633
95532890315	402607741907	62289451198	-236875048429
62678881454	-379579411070	-314029482460	-175106262326
250369178527	-9623821062	-123510255443	-1310466472
38117676779	392492935909	35210610278	113469390441
-307560097467	-34873784600	76471582039	378008759772
-451476943005	-113907652333	-201724891046	199607394108
39951677980	94695436468	655891013556	472342734602
50693373843	45033600173	-22688551154	20464484392
347553723356	-593135442713	194947044065	138440642418
73852982783	-40897519013	-41852396773	325637018310
346736092872	97749223458	296174872125	198462999980
46362643414	163614599131	-216347377208	345916551401
136639775707	237585899534	303815516355	-367766675740
-558115466920	155279073146	-90104765057	-459376405189
-17125221919	-85249226949	-79462786945	43017349691
-41476281979	-109230939013	137616575063	-67345596839
340787553401	228309232740	97197300635	-216343596307
208360109968	-284184399932	447688793182	603256424340
-300844324506	240416466735	175897596058	192019754791
-52504505037	-166662072814	168712861980	-91108465756
-275807747535	-178033218739	-7650789584	-258677872256
-296219549033	22960888058	-256503454481	426042066327
299263811838	-255043029829	316136691880	-88494733844
-122572875009	336004202929	245681188398	-181420699981
484497394714	-405667980252	-438773180450	-122766409888
-75231707174	-107113607282	100301913529	117168272819
-118505936459	442898819185	331952731237	-33696889761
-282991489922	-357744420120	106588427831	-87096753830
-587217849917	-228012934550	224827361274	-19423815219
2913776202	558109562640	67812784343	-580316855213
77012544004	601076336653	373852694423	486956639281
190526683224	173900709183	234858733836	-150023361136
17311169833	-181850827370	-406534518407	424400811758
358182653317	-205637958368	-37668387272	79176765243
-173558139901	134387589145	82562087482	48454180763
-116073733215	-51779210131	-161839854986	168475247021
-40460031263	-88464231948	6519607640	212566209839
-211346731649	-45457964566	330120120344	25914552749
-219328473858	-120390175791	23400716681	59583354679
124301144234	120724414206	-92926754042	43634738639
347455994773	-158332819159	46283141048	-202361878775
-84798462513	-37796657042	388129454636	190317500157
-6016918039	271494958628	-55101626572	-240048650980
-446833473102	-248422595471	-276808038735	10505477837
641393493623	-227802125507	-166394261545	394377365869
-255624719409	17515703725	136709480619	-35289008341
-24218814042	149693004212	-625021141940	-298098666987
-138398493143	-331015058463	-434085590512	-237183571455
66411881155	65250756082	-16647190738	-114030426869
156290717699	-111306450254	47672135161	-130373916267
14944774410	-356352613371	146088062083	70230778818
-196552853480	-499907800842	547654360053	189895739145
-50384458624	-130402126808	-48167471261	207064197599
-165305518524	158402930648	-328926534285	83286147716
18626998179	-27163305485	-234879761834	-67826550263
256025713379	-209008547879	22397852880	-579085310631
-325123738814	-454497886176	99342454491	-450121508267
-476163476620	164052491188	-107478516203	186110524991